

Project Report: Amazon Sales Data Analysis

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Overview

This project explores Amazon sales data to understand customer behavior, product trends, and sales patterns.

Using Python and data mining techniques, we cleaned and analyzed the dataset to uncover useful insights that can help businesses improve their marketing strategies, boost customer retention, and increase sales.

Team Members

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 Tools and Techniques

Python (pandas, scikit-learn, mlxtend)

Jupyter Notebook

Data Preprocessing (cleaning, encoding, normalization)

Machine Learning Models:

Classification (Decision Tree)

Clustering (K-Means)

Association Rule Mining (Apriori Algorithm)

 Steps We Followed

Data Preprocessing

Loaded the data from CSV

Handled missing values by dropping or cleaning incomplete columns

Removed duplicates to keep the data clean

Converted data types (like changing dates to datetime format)

Encoded categorical variables (turning words into numbers)

Normalized numeric features (scaling numbers to 0–1 range)

Model Building and Analysis

Classification: Predicted customer spending groups (low, medium, high)

Clustering: Grouped similar customers to discover hidden segments

Association Rules: Found frequently bought-together products for cross-selling

Key Findings

Identified valuable customer segments for better targeting
Discovered popular product combinations that could be bundled together
Highlighted top-selling categories and customer preferences
Achieved good model performance with simple, interpretable methods

Project Shortcomings

Limited dataset features (e.g., no customer demographics)
Time constraints limited model tuning and testing
Some simplifying assumptions in clustering and rule mining

Future Improvements

Expand dataset with more variables (customer age, location, etc.)
Apply advanced models like Random Forest or Gradient Boosting
Use real-time data for dynamic recommendations and predictions

Conclusion

This project shows how data mining can transform raw sales data into valuable insights. By applying simple, clear techniques, we were able to find patterns and trends that businesses can use to improve marketing, strengthen customer loyalty, and optimize product offerings.

Thank you for reviewing our work!